

BENJAMIN RIOT-BRETÊCHER

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EDUCATION

Ph.D - Atmospheric and Oceanic Sciences, McGill University 2022 -

Supervisor: [Prof. Yi Huang](#)

Relevant Coursework: Atmospheric Dynamics, Atmospheric Radiation, Numerical Methods, Waves & Stability

B.S - Atmospheric and Oceanic Sciences, Université du Québec à Montréal 2019 - 2022

PROFESSIONAL EXPERIENCE

Teaching Assistant 2023 -

McGill University

- [Introduction to Atmospheric Sciences - ATOC181](#) (Fall 2023, Fall 2025)
- [Science of Storms - ATOC184](#) (Winter 2024)

Visitor Student Oct 2024 - Jan 2025

[School of Atmospheric Sciences, Nanjing University](#)

Treasurer Aug 2023 - Aug 2024

Council of Atmospheric and Oceanic Sciences (CAOS)

Field Campaign Involvement 2022 - Present

McGill University

- [ECALOT Field Campaign](#) (Fall 2024)
- [WHAFFERS Field Campaign](#) (Winter 2025)
- [GHG Montreal Project](#) (Ongoing)
- [PONEX](#) (Ongoing)

Research Assistant May 2022 - Aug 2022

McGill University

Supervisor: [Prof. Yi Huang](#)

Research Assistant June 2021 - Aug 2021

Université du Québec à Montréal

Supervisor: [Prof. Jean-Pierre Blanchet](#)

PUBLICATIONS

- **B. Riot-Bretêcher**, Y. Huang, L. Liu, J. Gero, [Strengthening Atmospheric Greenhouse Effect at the Arctic North Slope of Alaska Site Evidenced by Long-Term Records of the Downwelling Longwave Radiance Spectrum](#), Journal of Geophysical Research: Atmospheres, 2025.

CONFERENCES AND TALKS

- [Joint FORUM-HAWC Workshop](#) - **Poster** - Montreal, QC, Canada - October 2025
- [Gordon Research Conference on Radiation and Climate](#) - **Invited Speaker** - Lewiston, ME, USA - July 2025
- [Gordon Research Seminar on Radiation and Climate](#) - **Speaker** - Lewiston, ME, USA - July 2025
- [Nanjing University - Atmospheric Sciences Seminar](#) - **Speaker** - Nanjing, China - December 2024
- [43rd Canadian Symposium on Remote Sensing](#) - **Attendee** - Québec City, QC, Canada - July 2022
- [Canadian Meteorological and Oceanographic Society](#) - **Attendee** - Remote - June 2021

AWARDS AND SCHOLARSHIP

- McGill Graduate Excellence Fellowship (2022 - 2028)
- Graduate Mobility Award - McGill University (Oct 2024 - Jan 2025) - \$ 3,800
- Canadian Meteorological and Oceanographic Society Undergraduate Scholarship (April 2021) - \$ 1,000

SKILLS

Technical Skills

MATLAB, Fortran, Python, Java, R, Bash, Linux, Radiative Transfer Model, Weather Balloon Launches, AERI Maintenance, Optimal Estimation Method

Mathematical Skills

Algebra, Calculus, Probability, Statistics

Certifications

- Data Security - Compute Ontario (June 2025)
- Introduction to Advanced Research Computing - Compute Ontario (June 2025)
- Introduction to Version Control Using Git - Compute Ontario (June 2025)
- Scaling Up HPC Workflows - Compute Ontario (June 2025)
- Career Essentials in Software Development by Microsoft and LinkedIn ([Certificate ID](#))
- Programming Foundation: Beyond the Fundamentals by LinkedIn ([Certificate ID](#))